

		Parent's signature	
Name	Index Number	CTG	

YISHUN JUNIOR COLLEGE
JC 2 PREMINARY EXAMINATION 2018

BIOLOGY

9744/02

HIGHER 2

29 AUGUST 2018

Wed 1400 – 1600

2 hours

Paper 2

Additional material: 3 pieces of writing paper



READ THESE INSTRUCTIONS FIRST

Write your name and CTG in the spaces at the top of this page and on all separate answer paper used.

Write in dark blue or black pen only.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the question paper.

Section B

Write your answers in the separate piece(s) of writing paper provided.

At the end of the examination, fasten all your work securely together.

This question paper consists of **19** printed pages and 1 blank page.

FOR EXAMINER'S USE	
Paper 1	/ 30
Paper 2	
Q1	/8
Q2	/9
Q3	/ 10
Q4	/7
Q5	/ 13
Q6	/ 7
Q7	/12
Q8	/10
Q9	/11
Q10	/13
Overall	/ 100

Section A

Answer **all** questions in this section.

Mark scheme abbreviations

; separates marking points

/ alternative answers for the same point

R reject

A accept (for answers correctly cued by the question, or by extra guidance)

AW alternative wording (where responses vary more than usual)

underline actual word given must be used by candidate (grammatical variants accepted)

max indicates the maximum number of marks that can be given

ora or reverse argument

mp marking point (with relevant number)

ecf error carried forward

I ignore

AVP alternative valid point

- 1 The cell theory was developed in the 1830s. At the same time, it was proposed that living things arose spontaneously from non-living materials. This theory of spontaneous generation" was later disproven, but the cell theory has stood the test of time to become widely accepted in the scientific community today.

(a) Outline the cell theory. [2]

any two:

1. living organisms composed of cells
2. cells form most basic (I smallest) unit of life
3. cells arise (I incorrect processes e.g. mitosis, cell division) from other cells

R endosymbiosis theory

Fig. 1.1 is an electron micrograph of part of a eukaryotic cell.

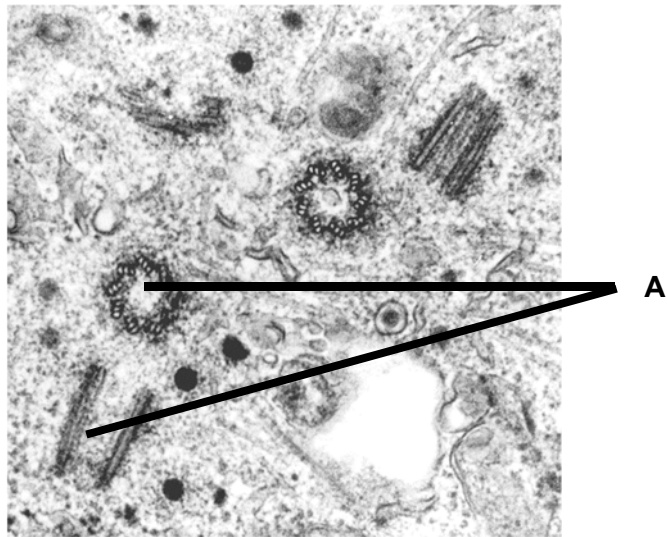


Fig 1.1

- (b) State how it is possible to deduce that Fig. 1.1 is a transmission electron micrograph and **not** a scanning electron micrograph. [1]

any one from:

- *idea that can see internal structures / can see organelles (e.g. centrioles, vesicles);*
- *cannot see surface contours / AW;*
- **A** not 3-D appearance
- **AVP**; e.g. ref. to small(er) depth of field;
- **I**: TEM is black & white / has different density of colouration

- (c) (i) Identify structure **A**. [1]

- Centrioles;
- **A**: a pair of centrioles
- **R**: singular

(ii) Describe the structural features shown in Fig. 1.1 that identify **A**. [2]

- A pair of hollow cylinders at right angle / perpendicular to each other;
- Each centriole consists of 9 triplets of microtubules arranged in a ring;
R: no mention of microtubules and ring arrangement

(iii) Cells such as that in Fig. 1.1 can divide by mitosis. [2]

Explain the role of structure **A** in mitosis.

two from

- organise microtubules / (function as) microtubule organising centre;
- (to), form spindle / assemble spindle fibres (in prophase); AW
- *ref. to* centriole pair / centrioles, at (both) poles during prophase;
R: if description is linked to incorrect mitotic stage;
- *ref. to* role in contraction / shortening of spindle fibres, at anaphase / to separate sister chromatids;
R: centrioles pull the chromatids apart;
- AVP ; e.g. make microtubules (as part of the centrosome)

[Total: 8]

- 2 Fig. 2.1 shows the structure of a triglyceride. Fig. 2.2 shows the structure of glycolipid.

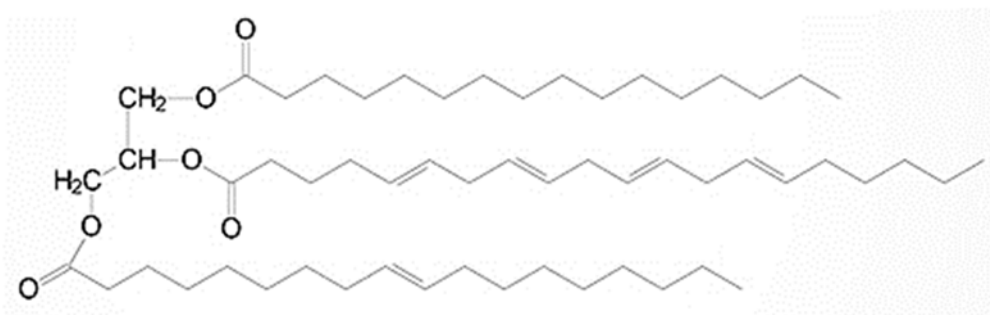


Fig. 2.1

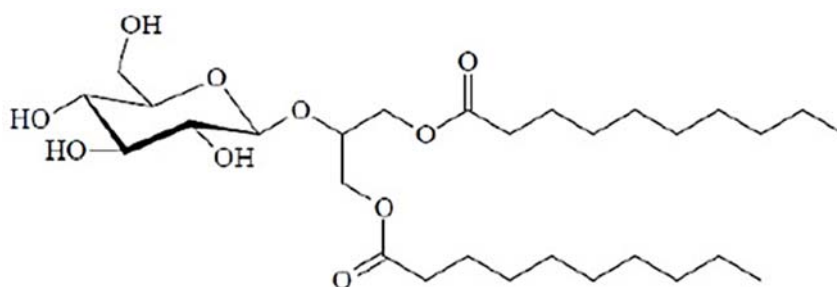


Fig. 2.1

- (a) Explain the difference in solubility between triglycerides and the products of their hydrolysis.[3]
1. Triglycerides are non-polar molecules that are insoluble in water. This is due to their incapability of forming hydrogen bonds with water molecules;
 2. **Glycerol and fatty acids** are the products of triglyceride hydrolysis;
 3. **Glycerol is an alcohol** that is soluble due to its ability (-OH groups in glycerol) to form hydrogen bonds with water;
 4. but **fatty acids** are insoluble due to their long hydrocarbon tails that are non-polar and hydrophobic.
- (b) Describe the roles of triglycerides and glycolipids in relation to their differences in molecular structure. [2]
1. Has long hydrocarbon chains, acting as energy store that yield more energy (38kJ per gram) than same mass of carbohydrates (17 kJ per g);
 2. Releases more metabolic water during fat oxidation/ respiration;
 3. Lipid molecule with a carbohydrate component and act as a recognition site for cell adhesion.

(c) Suggest, in terms of membrane structure, how plants can be tolerant to freezing. [4]

1. Increase in the amount of unsaturated fatty acid chains – unsaturated fatty acids;
2. have kinks in their fatty acid tails, thus preventing close packing of the molecules and decreasing the amount of interaction between adjacent fatty acid chains;
3. Have shorter chain length of the fatty acids;
4. reduces the tendency of the hydrocarbon tails to interact with one another by weak hydrophobic interactions between the fatty acid tails;
5. Presence of cholesterol molecules that orient themselves in the bilayer;
6. Keeps the membrane fluid by preventing the hydrocarbon fatty acid chains of the membrane lipids from binding to one another./helps to regulate cell membrane fluidity by resisting changes in membrane fluidity that can be caused by changes in temperature.

[Total: 9]

- 3 High fructose corn syrup, made from maize, can be used as a replacement for sucrose to sweeten food and drink products.

Commercial production of high fructose corn syrup involves the enzyme glucose isomerase, extracted from bacteria.

Fig. 3.1 shows the results of an investigation into the effect of glucose concentration on the rate of reaction catalysed by glucose isomerase.

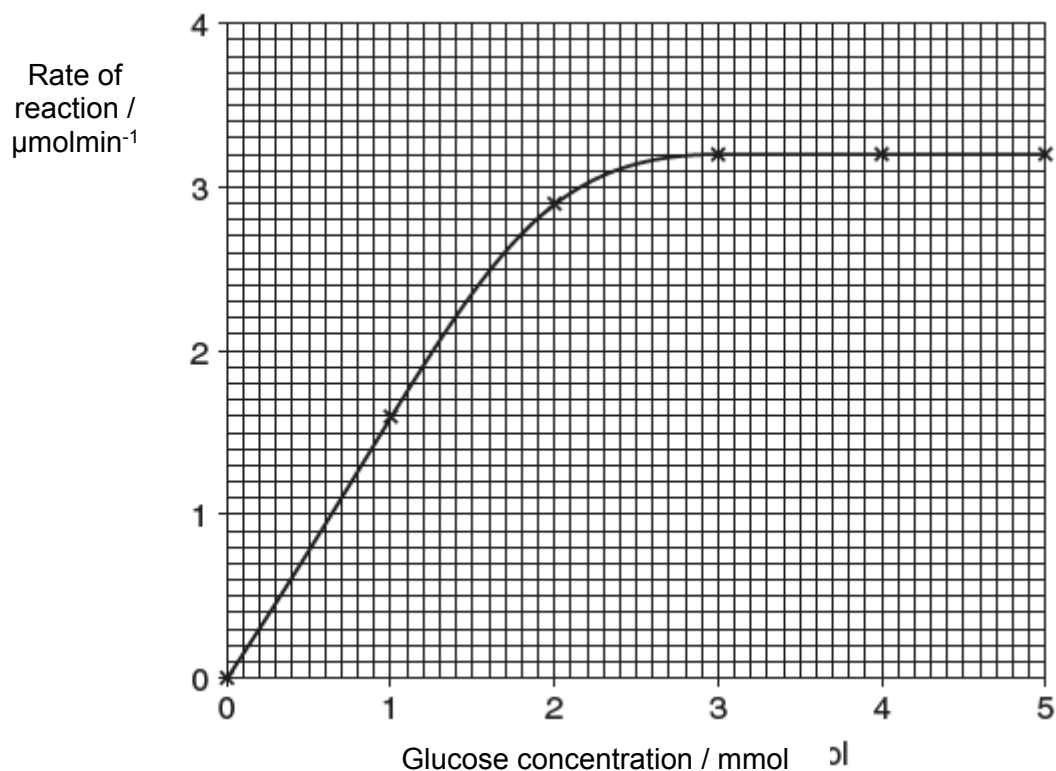


Fig. 3.1

- (a) Use Fig. 3.1,
- (i) state the lowest substrate concentration to give the maximum rate of reaction, V_{max} . [1]
 1. 2.9 mmol; **A** 2.8–3.0 mmol
R: no unit
 - (ii) determine the Michaelis-Menten constant, K_m . [1]
 2. 1 mmol;
A: 0.95–1.05 mmol
R: no unit

- (b) The glucose isomerase used in the production of high fructose corn syrup is extracted from a strain of a bacterium, *Thermus thermophilus*, which is found in hot springs. The enzyme has an optimum temperature of 95 °C.

Suggest **and** explain the advantages of using glucose isomerase from *T. thermophilus* to produce high fructose corn syrup, rather than using glucose isomerase that has an optimum temperature of 37 °C. [3]

look for ora throughout if describing the other enzyme with low optimum three from

1. *idea of* can use high(er) temperatures for process;
2. increased temperature increases, number of collisions (between enzyme
3. and substrate) / number of ES complexes (formed) ; **R** no mention of increased temperature;
4. more product / high(er) rate of reaction; AW
5. less prone to denaturation; **A** won't denature
6. described in terms of loss of active site
7. more stable / lasts longer;

A thermostable

A reused over and over / need not be replaced (during course of production) / (allows) automation

I: temperature resistant

The commercial production of high fructose corn syrup uses immobilised glucose isomerase.

Fig. 3.1 shows the effect of temperature on the activity of immobilised glucose isomerase compared to glucose isomerase free in solution.

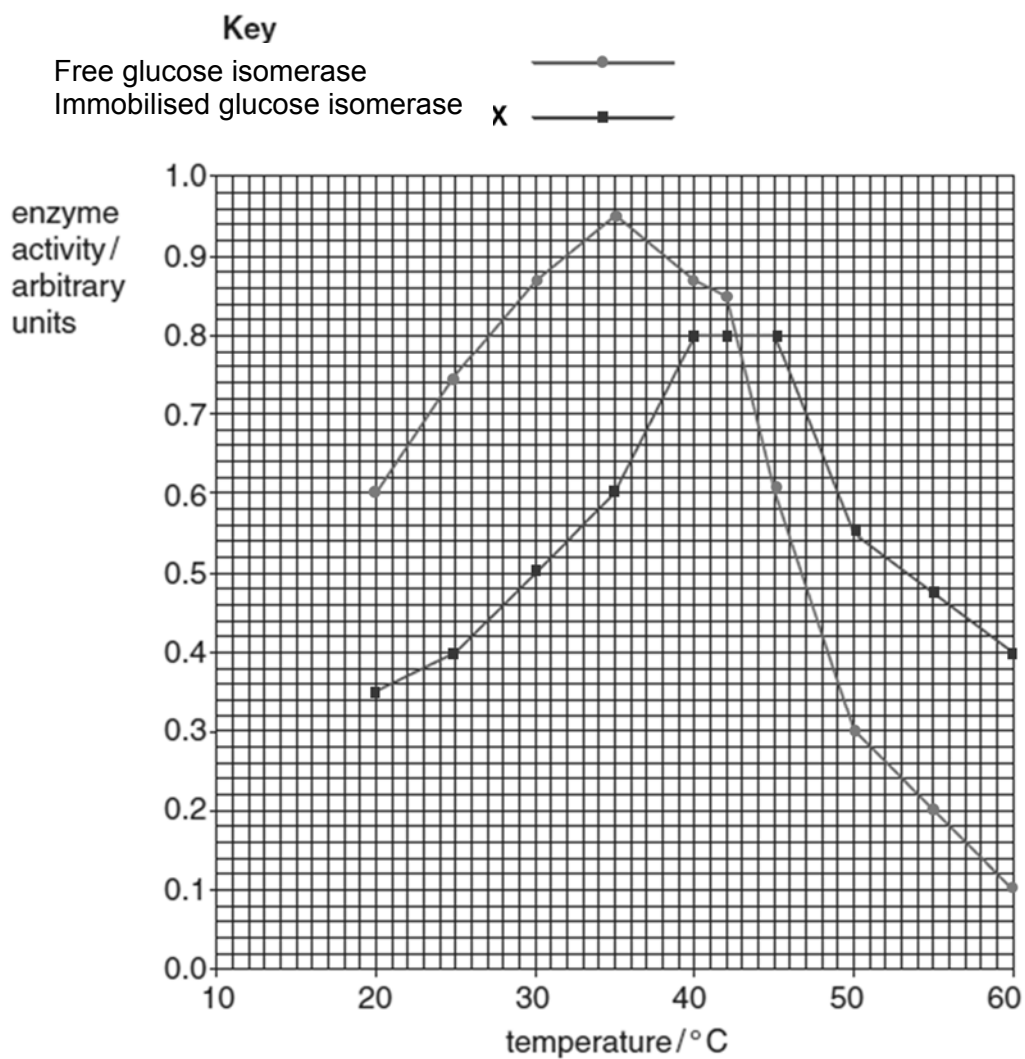


Fig. 3.2

(c) With reference to the Fig. 3.2,

- (i) describe and explain the effect of temperature on the activity of free glucose isomerase. [3]

Describe

- As temperature increases from 20 to 35 °C, the rate of reaction of free glucose isomerase increases from 0.60 to 0.95 A.U. due to the increased kinetic energy of the glucose and glucose isomerase molecules;

Explain

- These molecules move faster colliding with one another in correct orientation to form more glucose isomerase-glucose complexes;

Describe

- The increased rate of reaction increases until it reaches its maximum rate of 0.95 A.U. at the optimum temperature of 35 °C;

Explain

- Most glucose isomerase-glucose complexes formed;

Describe

- Beyond the optimum temperature of 35 °C to 60 °C, rate of reaction decreases from 0.95 to 0.1 A.U.

Explain

- intramolecular bonds (reference to hydrogen bonds, hydrophobic interactions) are broken and three-dimensional conformation of active sites in glucose isomerase is altered such that glucose can no longer fit. **A** enzyme is denatured

[1 mark each, max 3 marks]

[minimum 1 describe + 1 explain]

[Max 2 marks if all description or explanation]

- (ii) describe and explain the differences between the activity of free and immobilized glucose isomerases up to 40 °C. [2]

Description

- From 20 to 40 °C, activity of immobilized glucose isomerase is lower than free glucose isomerase (reference to any suitable data range); OR ref to data point that immobilised enzyme has lower activity for all temperature up to 40 °C

[max 1 mark]

Explanation

- Immobilized glucose isomerase less easily to encounter glucose/ less able to move and collide successfully with glucose leading to fewer glucose isomerase-glucose complexes formed; I immobilized enzymes do not get denatured easily than free enzymes

[max 1 mark]

[Total: 10]

- 4 Fig. 4.1 shows information about the movement of chromatids in a cell that has just started metaphase of mitosis.

key

--- distance between chromatids

— distance between each chromatid and the pole to which it is moving

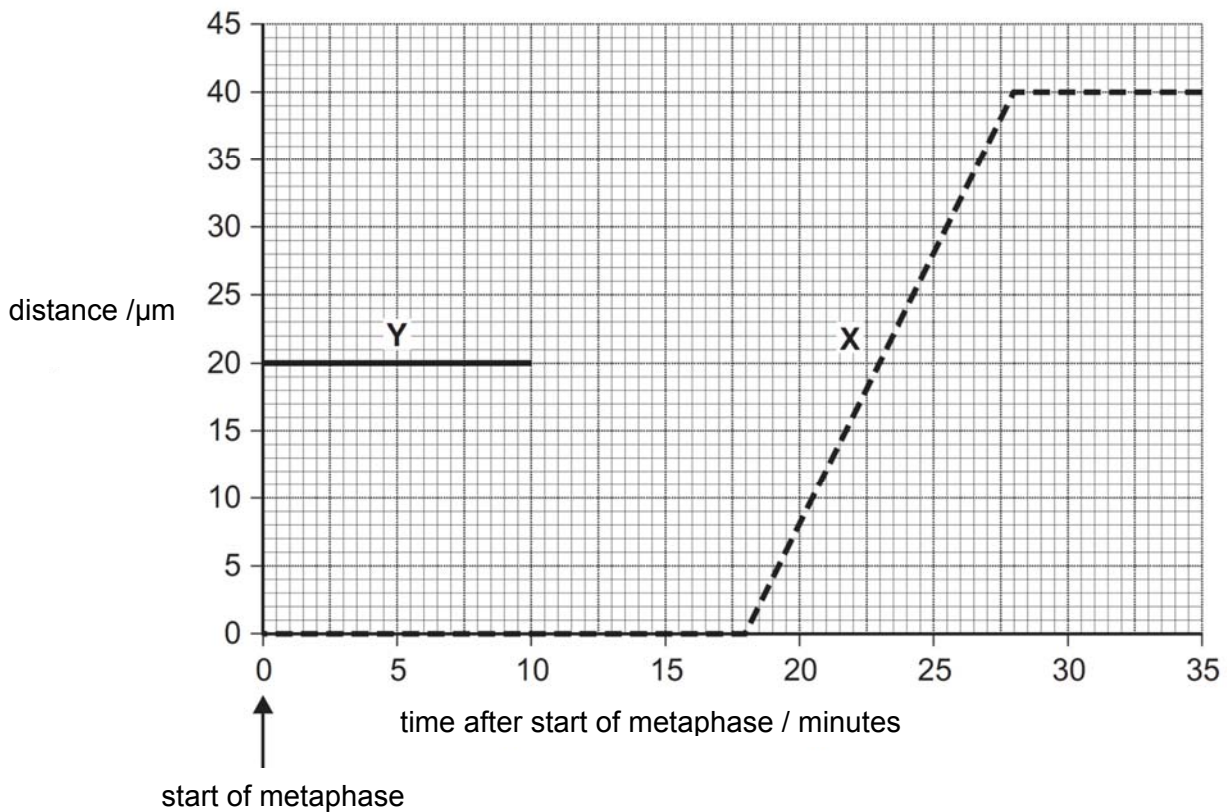


Fig. 4.1

With reference to Fig. 4.1,

- (a) (i) state the duration of metaphase in the cell. [1]

• 18min ;

- (ii) complete line Y on the graph. [1]

• Horizontal until 18 minutes, then decreases as straight line to 0 μm at 28 minutes;

(iii) account for your answer in (a)(ii). [3]

- Chromosomes align singly at the metaphase plate during metaphase of mitosis as distance between each chromatid and the pole remains constant at 20 μm till 18 minutes; **R** no ref to chromosome behavior / just stating the stage of mitosis
- Sister chromatids separate at the centromere to become daughter chromosomes and migrate towards the opposite poles in anaphase as distance between each chromatid and the pole starts to decrease from 20 μm to 0 μm at 28 minutes.
- Each chromatid / daughter chromosome did not move / remain at the pole in telophase.

**quoting of data is necessary to support the answers above.*

The movement of chromatids is dependent on spindle fibres, which are made up of many tubulin subunits. Spindle fibres are lengthened at one end during mitosis by the polymerisation of tubulin subunits through GTP hydrolysis.

A drug, eribulin, is known to prevent the polymerisation of the tubulin subunits.

(b) Suggest and explain the effect of eribulin on the behaviour of chromosomes in mitosis. [2]

- Kinetochore microtubules cannot attach to the kinetochores at the centromeres of the chromosomes.
- Cells cannot progress through metaphase, so that chromosomes cannot align singly at the metaphase plate.
- Sister chromatids could not separate / remain attached in anaphase / unequal (random) separation of chromosomes;
- **A** spindle fibres are not formed / microtubules are not lengthen (from centrioles);

[any 2]

R no ref to behavior of chromosomes

[Total: 7]

- 5 Erythropoietin, also known as EPO, is a large glycoprotein synthesised by specialised cells in the kidney. These cells are very sensitive to changes in oxygen concentration in the blood passing through the kidney and respond to a low oxygen concentration by increasing the synthesis of EPO.

EPO acts at the surface of particular target cells, such as cells in the bone marrow. These bone marrow cells are stimulated to produce red blood cells.

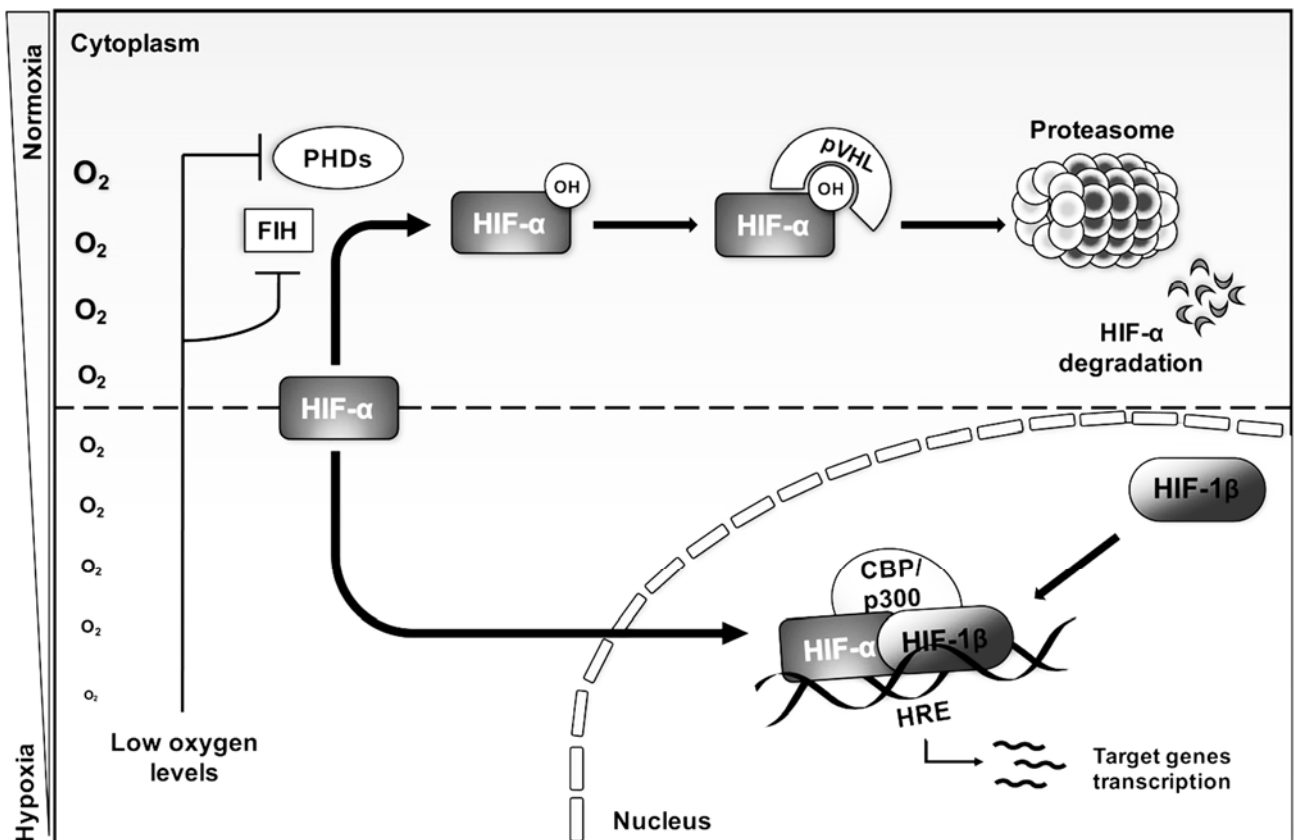
- (a) State precisely the location(s) where synthesis of the RNA(s) involved in EPO synthesis is / are located in the specialised cells in the kidney.

- Synthesis of rRNA (with ribosomal proteins form ribosomal subunits required for translation): **nucleolus**
- Transcription of gene coding for EPO / synthesis of mRNA encoding EPO: **in nucleoplasm**
- Transcription of gene encoding tRNA / synthesis of tRNA: **in nucleoplasm**

[All correct for 1 mark]

[1]

Fig. 5.1 shows the effect of varying oxygen concentration on EPO gene expression.



Biomedicines Review

Legend -

- CBP: CREB binding protein
- HIF: Hypoxia Inducible Factor
- HRE: Hypoxia response element

Fig. 5.1

- (b) A low oxygen concentration (hypoxia) also leads to an increase in the quantity of mRNA in the specialised cells in the kidney.

With reference to Fig. 5.1 and your own knowledge of transcription, explain how the increase in the quantity of EPO mRNA is achieved.

At low oxygen concentration / hypoxia,

1. **HIF- α / hypoxia inducible factor** is transported through the nuclear pore into the nucleoplasm / nucleus;
2. where it **forms a complex with HIF-1 β and CBP (CREB binding protein) /p300** that binds to the **enhancer sequence, HRE / hypoxia response element**;
3. **DNA-bending protein** brings the **HIF- α , HIF-1 β and CBP complex or activator** complex closer to the promoter site;
4. Core promoter **more accessible** to basal transcription factors & RNA polymerase;
5. Activator complex / HIF- α , HIF-1 β and CBP complex binds to mediator protein, triggering assembly of transcription of initiation complex, leading to an increase in transcription of EPO gene / increased synthesis of EPO mRNA.

OR

6. Activator complex / HIF- α , HIF-1 β and CBP complex can recruit HATs to change chromatin structure at the regulatory sequence and promoter to increase access of basal transcription factors and RNA polymerase to promoter site, leading to an increase in transcription of EPO gene / increased synthesis of EPO mRNA.

[1 mark each, max 4]
[4]

- (c) Protein synthesis accounts for a large proportion of the energy budget of a cell and therefore requires tight regulation.

Describe how translational control regulates gene expression.

Regulation of translation of all mRNAs

1. Translation initiation factors may be present or absent / activated or inactivated to start or prevent initiation of translation of all mRNAs in a cell;
2. This allows the cell to shut down translation if environmental conditions are poor or until the appropriate conditions exist.

OR

Regulation of translation of specific mRNAs

1. Initiation of translation is blocked by repressor proteins that bind to the 5'UTR of mRNA, preventing the binding of the small ribosomal subunit;
2. Initiation of translation is prevented because mRNAs lack poly(A) tails of sufficient length.

[2]

- (d) At high oxygen concentrations, HIF- α is degraded inside the proteasome. Suggest an advantage of degrading HIF- α at high oxygen concentrations.

- Can recycle amino acids for synthesis of other proteins;
- Prevent deleterious effects that may arise if there is an accumulation of unwanted protein in the cytoplasm.

[Any 1 for 1 mark]

[1]

- (e) EPO cannot pass through the cell surface membrane to enter the bone marrow cell.

Suggest **one** reason why this is so.

- EPO is too large to pass through the cell surface membrane;
- EPO is hydrophilic / water soluble, and is unable to cross the hydrophobic core of the phospholipid bilayer;
- EPO is hydrophilic / water soluble and the bone marrow cell surface membrane lacks a specific transport protein for it;

[Any 1 for 1 mark]

[1]

- (f) As part of an investigation As part of an investigation into the body's response to EPO, a group of healthy young men were given injections of EPO every day for four weeks.

The haemoglobin (Hb) concentration for each subject was measured at the start of the investigation and then at intervals of one week for the next ten weeks. The first measurement was taken two weeks before the first EPO injection was given.

Fig. 5.2 shows the mean results for the subjects.

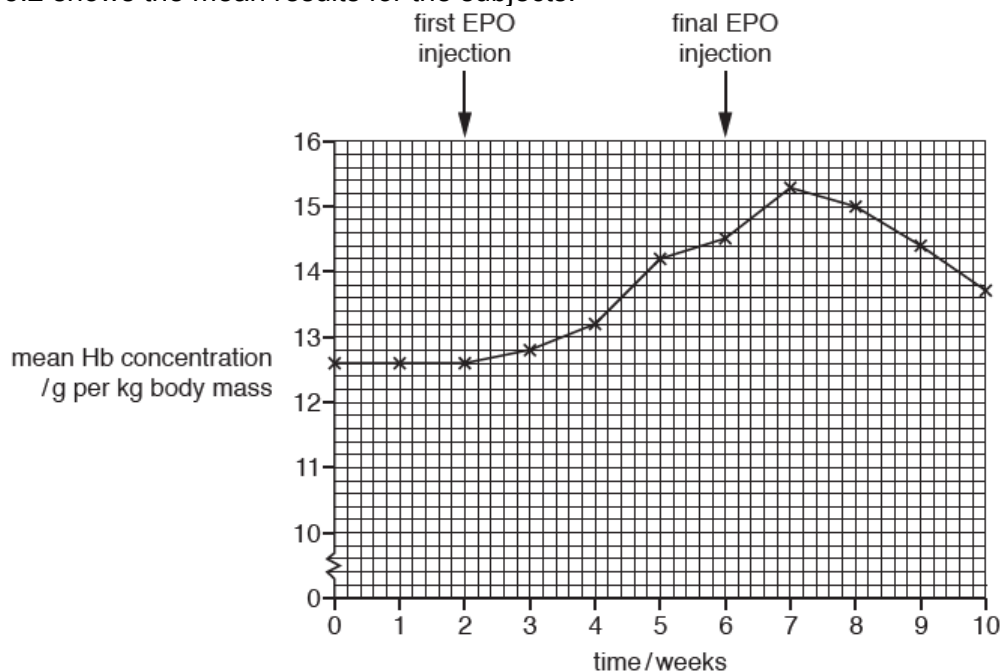


Fig. 5.2

Describe the results shown in Fig. 5.2 and suggest explanations for these results.

max 3 if all description or all explanation

- **A:** Hb for haemoglobin and Hb concentration for mean Hb concentration;

Constant trend

- **Describe:** Hb concentration, remains constant / of 12.6 g kg^{-1} , for first two weeks (of investigation) / up to start of injections;
- **Explain:** idea of regulation;
 - e.g. sufficient oxygen so no requirement for increased EPO

Increase then decrease description

- **Describe:** (then) increase in Hb concentration (from week 2) for 5 weeks / AW, then decrease (for last three weeks / to week 10);
- **Describe:** data quote / manipulated data, to support;
 - e.g. increase from 12.6 g kg^{-1} (week 2) to 15.3 g kg^{-1} (week 7)
 - increases by 2.7 g kg^{-1} (to week 7)
 - decrease from 15.3 g kg^{-1} (week 7) to 13.7 g kg^{-1} (week 10)
 - decreases by 1.6 g kg^{-1} (to week 10)

explanation for increase

- **Explain:** EPO increases production of red blood cells that contain Hb / AW;

Explanation for decrease

- **Explain:** red blood cells, short life span / die ;
- **Explain:** cell signalling stops / (target / bone marrow) cells no longer stimulated / AW;

A: EPO is degraded / AW

increase after injections stop

- **Describe:** Hb concentration increases for 1 week after injections have finished ;
- **Explain:** idea of, time delay for red blood cell production to stop / time for immature red blood cells to mature and be released into blood stream ;

AVP:

- e.g. steady increase as time required for, mitosis / cell proliferation / differentiation into red blood cells / production of haemoglobin;
- contributory factor for increase may be, accumulation / increased concentration, of EPO with injections;

[4]

[Total: 13]

- 6 Normal muscle fibres can be generated if adult mesenchymal stem cells, without the genetic defect that causes Duchenne muscular dystrophy (DMD), is delivered into the patients' muscles. Mesenchymal stem cells can give rise to many types of cells including bone, cartilage, lung and muscle cells. They are known as adult stem cells.

(a) Describe **two** similarities between mesenchymal stem cells and cancer cells. [2]

- Active telomerase activity that maintain the telomere length;
- Unspecialized cells with no specific function ;
- Capable of long-term cell and nuclear division;
- produce genetically identical cells ;
- AVP ;

R: if no supporting details / elaboration / merely stating of terms

[any 2]

Fig. 6.1 shows a possible approach for DMD gene therapy in humans to allow for the expression of full-length dystrophin protein. Dystrophin protein supports muscle fibre strength.

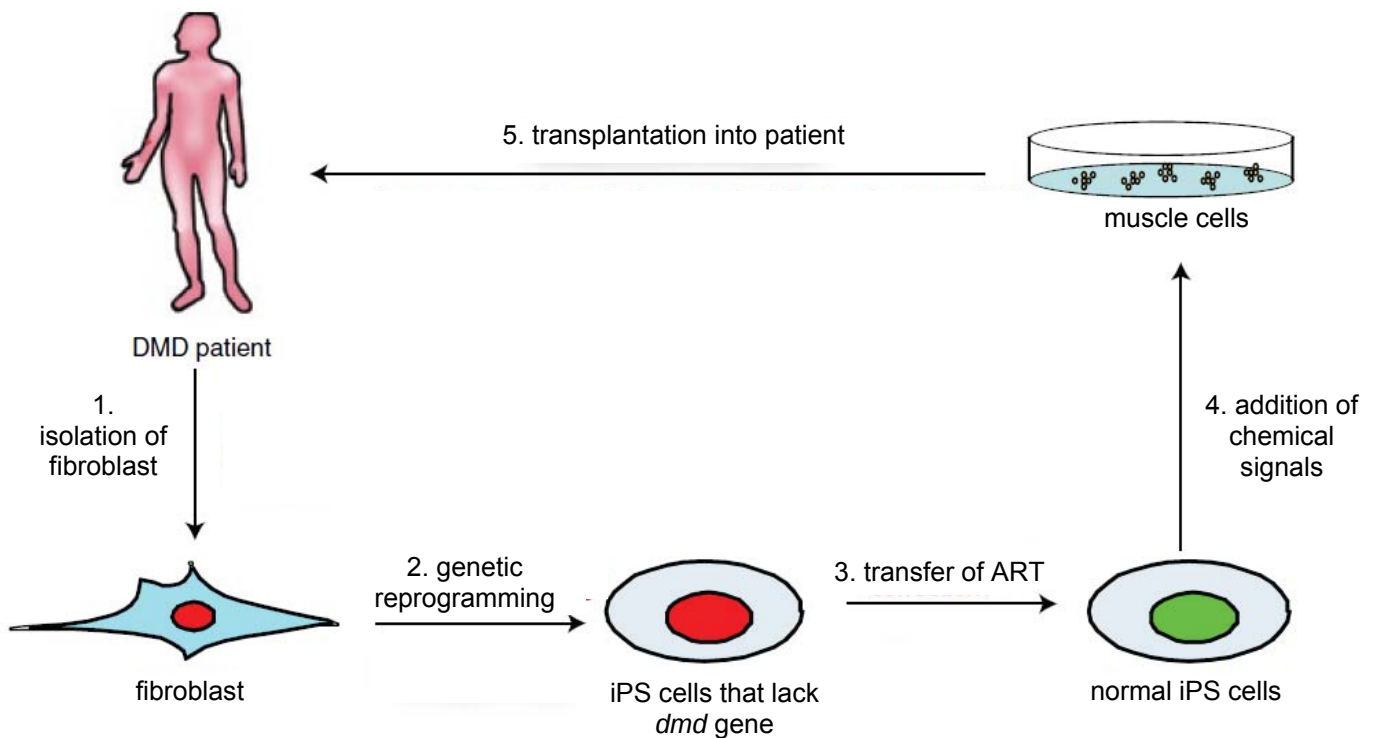


Fig. 6.1

(b) Explain why it is preferable to isolate fibroblast from the the DMD patient. [2]

- it will not induce immune rejection when transplanted back into patients;
R: tissue rejection
A: cell rejection / immune-rejection / immune response
- fibroblasts contain the same self-antigens as the patient;
- immune cells can recognise fibroblasts as self;
A: reduce ethical concern as no embryos are destroyed in the process;

(c) Upon transplantation into the patient, suggest disadvantages of this approach. [3]

1. Efficiency of transfer of muscle cells into patient maybe low;
2. Cells may not produce enough DMD protein;
3. The risk of stimulating the immune system leads to rejection by the host;
4. Uncontrolled cell division → leading to formation of tumour / cancerous growth;
5. Replacement of somatic cells → not a permanent solution to DMD;
6. Invasive procedure → more complications during post-surgery;

[Total: 7]

7 Fig. 7.1 shows the process of binary fission in bacteria.



Fig. 7.1

(a) (i) Briefly describe the events occurring in binary fission.

1. DNA replication occurs **bidirectionally**, from an **origin of replication**, which opens up into a **single replication bubble**;
2. The replication bubble **separates the DNA double helix**, each strand acts as a **template for synthesis of a daughter strand**; by **semi-conservative replication**;
3. the **cell elongates** and separates the daughter DNA molecules that are attached to the cell membrane;
4. **invagination of the plasma membrane**; and **cell wall forms during cytokinesis**, forming two **genetically identical daughter cells**;

[Max 3 for 3 marks]

[3]

(ii) Contrast binary fission and the process of cell division in an animal cell.

Feature	Binary fission	Cell division
1. Type of division	It is cell division.	It is a nuclear division stage (Mitosis) followed by cytokinesis
2. Importance/significance of process/ Function	The <u>process occurs for cells to reproduce asexually</u> .	The process occurs for the <u>growth and repair</u> of worn-out parts of the body of a <u>multicellular organism</u> and is the <u>basis of asexual reproduction</u> for a <u>unicellular organism</u> .
3. Attachment of DNA/ chromosome	In binary fission, the bacterial ' <u>chromosome</u> '/ DNA is <u>attached to the cell membrane during DNA replication</u> .	In the mitosis phase of cell division, <u>chromosomes are attached to spindle fibres</u> at their centromeres.

Feature	Binary fission	Cell division
4. Structures involved in process	Binary fission does not require these structures such as the <u>spindle fibres, centrioles/ microtubule-organising centres/ centrosomes, centromeres and kinetochores</u>	In the mitosis phase, structures such as <u>spindle fibres, centrioles/ microtubule-organising centres/ centrosomes, centromeres and kinetochores</u> are required.
5. Occurrence of DNA replication	DNA replication in bacteria occurs <u>during</u> binary fission	DNA replication occurs <u>prior to mitosis</u> so that at prophase, <u>chromosome appears as a double structure</u> .
6. Separation of DNA/ chromosome	As the bacterial <u>cell elongates</u> after DNA replication, each circular <u>DNA strand which is attached to the cell membrane separate</u> .	<u>Sister chromatids</u> of each chromosome <u>separate</u> and move towards opposite poles during anaphase (and each is a chromosome). <u>Nuclear envelope reforms</u> around the set of chromosomes at each pole during telophase.
7. Formation of daughter cells	<u>New cell wall and membrane grow between the DNA molecules</u> so that two identical daughter cells are formed.	<u>Cytokinesis occurs after mitosis</u> where the cytoplasm divides into two parts separating the two identical nuclei into two daughter cells.
8. Appearance of DNA	<u>Circular DNA is unwound from each other</u> with the help of enzyme <u>gyrase which removes positive supercoiling</u> as replication takes place.	Each <u>linear DNA</u> is packaged into chromatin thread which further <u>condenses</u> during prophase of mitosis <u>to form chromosome</u> .

[Any 2 differences for 2 marks]

[2]

Recent findings suggest that the mucus lining the human digestive tract is loaded with bacteriophages. These bacteriophages contain proteins which bind to the carbohydrate residues of mucoproteins as shown in Fig. 7.2. It has been proposed that these bacteriophages can play an important role as part of the body's defense system.

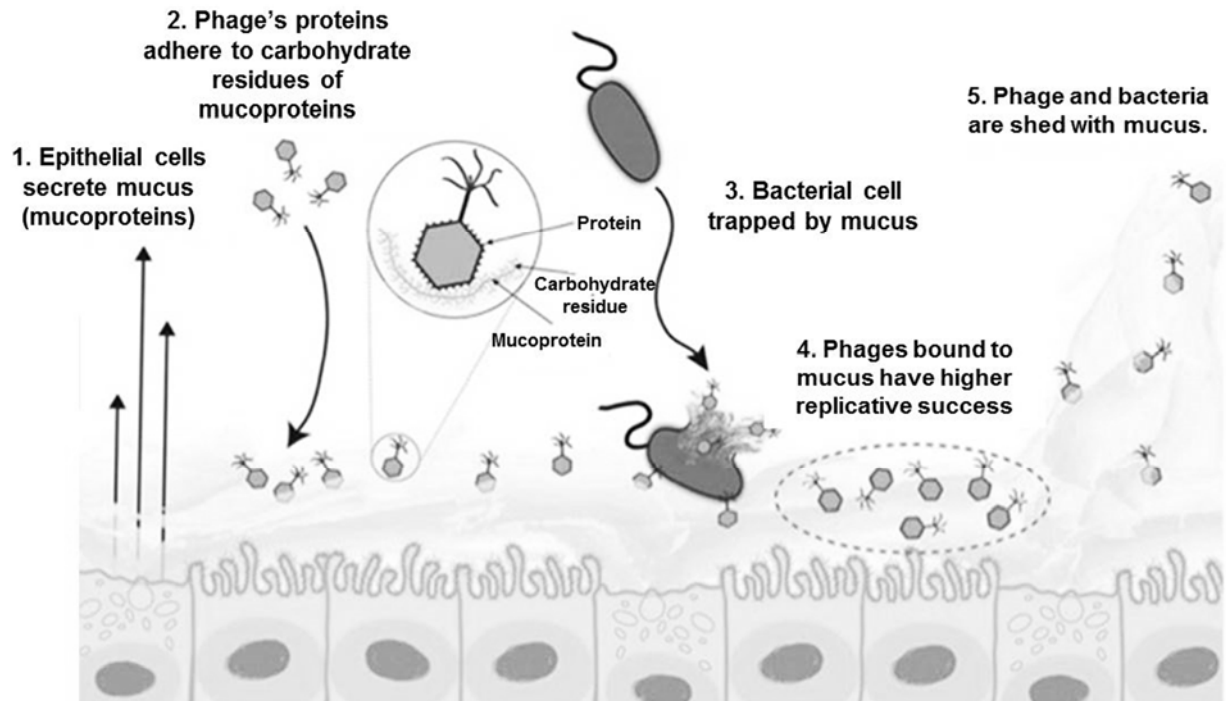


Fig. 7.2

(b) Describe the life cycle of the bacteriophages shown in Fig. 7.2.

1. The bacteriophages use their tail fibres to bind to specific receptors on host bacteria;
2. Phage-coded enzyme, lysozyme, digests part of the cell wall of host bacterium/ and viral DNA/ genome is injected;
3. Ref. to transcription and translation/ expression of phage proteins by using host RNA polymerase and ribosomes;
4. Ref. to replication of phage DNA using host DNA polymerase;
5. Newly synthesized bacteriophage proteins and DNA assembled into new phages and are released from the bacteria by osmotic lysis, killing bacteria in the process;

[max 4 for 4 marks]

[4]

- (c) Many other viruses may also be found within the human body e.g. the Human Immunodeficiency Virus (HIV).

Compare the lysogenic life cycle of a temperate bacteriophage and the life cycle of a HIV.

Similarity (1 mark each):

1. Both life cycles involve integration of viral DNA into host cell genome;
2. Both life cycles involve a period of dormancy;
3. Both consist of proteins which bind to specific host cell receptors;

Difference (1 mark per difference):

Feature	Lysogenic life cycle	HIV life cycle
4. Mode of adsorption / attachment onto host cell surface	Use tail tip to bind to specific receptors on bacterial cells	Use gp120 to bind to CD4 receptors on Helper T cells
5. Viral content that enters host cell upon penetration	Only DNA is injected into host cell /Capsid left outside host cell	Capsid and viral RNA enters host cell
6. Integration of viral genome	Viral DNA integrated into host cell genome directly	Viral RNA needs to be reversed transcribed to viral DNA before integration into host cell genome
7. Type of host cell	Occurs in bacteria	Occurs in T helper cells/ cells with CD4 receptors
8. Death of host cell	Do not result in death of the host cell	HIV may result in death of host cell

[At least 1 similarity and 1 difference, max 3 marks]
[3]

[Total: 12]

- 8 In an experiment involving deer mouse, pure breeding pink-eyed mice with wild-type fur was crossed with pure breeding dark-eyed albino mice.

The resulting progeny all had wild-type fur and dark eyes. These F_1 mice were then crossed with pink-eyed albino mice. The results are shown in Table 8.1. It was difficult to distinguish between mice that are dark-eyed albino and pink-eyed albino, so these two phenotypes were counted together (F.H. Clark. 1936. *Journal of Hereditary* 27:259-260).

Table 8.1

Phenotype	Number of progeny
Dark-eyed, wild fur (recombinant)	0
Pink-eyed, wild fur (parental)	63
Dark-eyed, albino (parental) Pink-eyed, albino (recombinant)	71

The calculated X^2 value is 32.8. Table 8.2 shows part of the table of probabilities for the chi square test.

Table 8.2

number of degrees of freedom (v)	distribution of X^2		
	probability		
	0.1	0.05	0.01
1	2.71	3.84	6.64
2	4.60	5.99	9.21
3	6.25	7.82	11.34
4	7.78	9.49	13.28

- (a) Explain the conclusion drawn from the Chi square test. [3]
- Difference between observed and expected results is significant
 - Because $P < 0.05$
 - Does not conform to the expected phenotypic ratio of 1:1:1:1 / 1:1:2
 - The difference not due to chance. Hence the two genes do not segregate independently
- (b) (i) Using D/d to represent alleles for eye colour and A/a to represent alleles for fur coat colour, explain the result of the F_1 cross in the deer mouse experiment using a genetic diagram. [4]

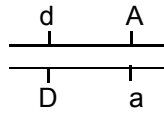
Key: D represents the dominant allele for dark-eyed.
 d represents the recessive allele for pink-eyed.
 A represents the dominant allele for wild-type fur.
 a represents the recessive allele for albino

*F*₁ Cross:**Phenotype of parents:**

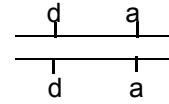
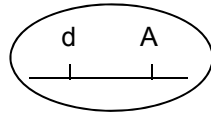
dark-eyed, wild type fur

x

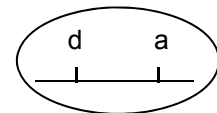
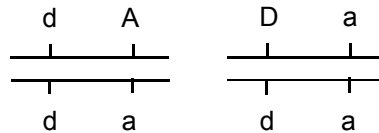
pink-eyed, albino

Genotype of parents:

x

**Gametes produced (n):**

x

**Genotype of offspring:****Phenotype of offspring:**pink-eyed
wild-type furdark-eyed
albino

Parental type

(ii) How would you explain the experimental ratio?

[3]

- The difference between the experimental ratios obtained and the expected ones may be due to the tightly linked genes for eye colour and fur.
- The alleles of these two characteristics are located close to each other on the same chromosome;
- Linked genes tend to be inherited together;
- Since there is complete linkage, minimal recombinant gametes formed as no crossing over can occur.
- Almost all parental phenotypes of ratio 1: 1 obtained due to complete linkage.

[Total: 10 marks]

- 9 ATP and NAD both play important roles in respiration. Both compounds are nucleotides. Fig. 9.1 represents the molecular structures of ATP and NAD.

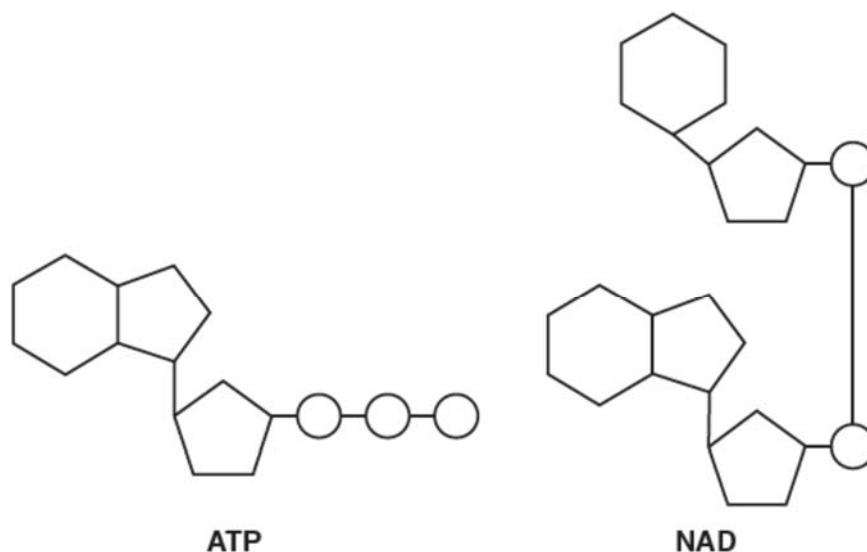


Fig. 9.1

- (a) Using Fig. 9.1, compare the structures of ATP and NAD. [3]
- both have ribose (sugars) ; **R**: ribulose
 - ATP has 1, ribose/ pentose/ sugar, NAD has 2 ; I ref. to additional hexose
 - both have, adenine/ purine (base) ; I adenosine
 - NAD has, nicotinamide/ pyrimidine (base) ;
 - ATP has 3 phosphates, NAD has 2 ;
- (b) Name the type of chemical reaction by which ATP is made during the Krebs cycle. [1]
- substrate-linked/ substrate-level, phosphorylation ; I condensation reaction
- (c) Outline the roles of NAD in the cytoplasm of a cell. [2]
- hydrogen, carrier/ acceptor;
A: gets reduced or gains H / H^+ and electrons
R: H_2 / hydrogen molecules
 - (acts as a) coenzyme;
A enables dehydrogenases to work ref. to glycolysis /respiration in anaerobic conditions;
A: anaerobic respiration
I: aerobic

- (d) An experiment was carried out to investigate the effect of epicatechin on mitochondrial respiration in mice. Epicatechin is a naturally occurring compound in cocoa beans and so is present in chocolate.

Two groups of mice, group A and group B, were used in this experiment.

- Group A was given water containing epicatechin, twice a day for 15 days.
- Group B was given water without epicatechin, twice a day for 15 days.

After 15 days, the structure of mitochondria from striated muscle cells in both groups of mice was examined.

The surface area of the inner membrane of the mitochondria was divided by the surface area of the outer membrane to obtain a ratio for each mouse.

Table 9.1 shows the mean ratios for the two groups of mice.

Table 9.1

group	mean ratio
A	2.0 : 1
B	1.7 : 1

The mice in group **A** were able to exercise longer than the mice in group **B**.

With reference to Table 9.1, explain why the mice in group **A** were able to exercise for longer than the mice in group **B**. [5]

Max 4 of:

group A (accept ora for group B throughout) accept 'they' = group A

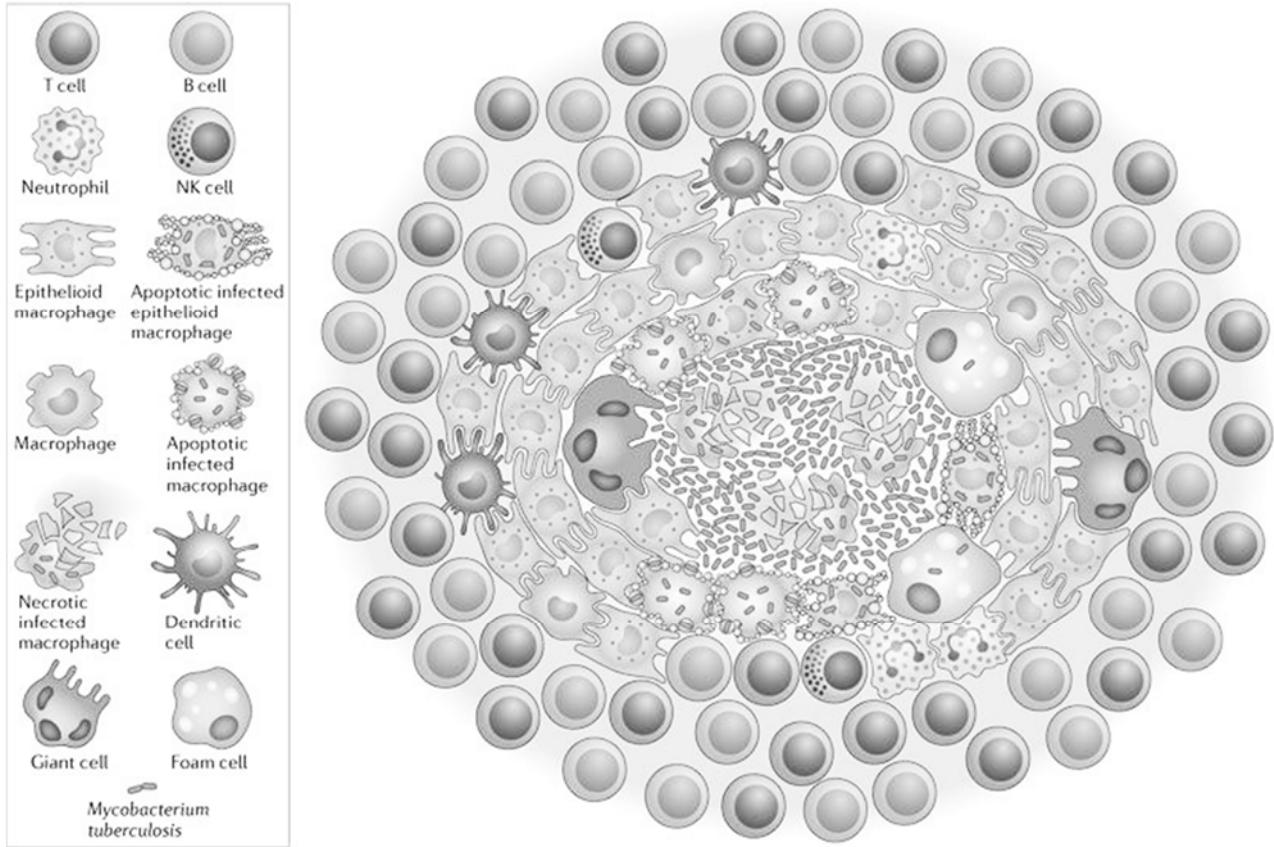
1. higher ratio;
2. larger / more, inner membrane / cristae (than **B**) ;
3. more, ETCs / cytochromes / ATP synth(et)ase / stalked particles ; I ATPase
4. oxidative phosphorylation occur ;
5. more ATP produced ;
6. muscles can contract for, longer / more time / without getting tired ; I exercise longer
I: muscles contract faster
7. AV ; e.g. chemiosmosis or detail thereof: H⁺ move, down gradient / through ATP synth(et)ase
I: ATPase

If B and A switched round penalise once only

[Total: 11]

- 10 Tuberculosis kills 2 million people worldwide every year. One-third of infected people have the latent disease. Latent TB (LTBI) is characterised by the formation of granulomas in the lung.

Fig. 10.1 below shows a diagram of a granuloma during LTBI.



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Fig. 10.1

(a) With reference to Fig. 10.1, describe how LTBI occurs.

1. This occurs because alveolar macrophages are unable to kill the bacteria they have engulfed due to the bacterium inhibiting the fusion of phagosome with lysosome;
2. Thus, lysosomal enzymes are thus not available to break down the bacteria ;
3. *M. tuberculosis* survives and continue to multiply inside the macrophages ;
4. infected cell secretes cytokines to attract other immune cells such as T cells and B cells ;
5. Results in formation of a granuloma where macrophages, T cells, and B cells form a wall / cluster around the undestroyed bacteria bacilli at the site of infection (lungs);
6. Necrosis of infected macrophages occurs in the core of the granuloma, where bacterium is released in the cavity, allowing it to replicate outside of the macrophages.

[max 4]
[4]

- (b) Streptomycin was the first antibiotic used to treat TB

During the first few years after the introduction of streptomycin treatment, an increasing number of *Mycobacterium tuberculosis* bacteria developed resistance to streptomycin.

- (i) Outline how this happened.

1. Incomplete treatment where dose of antibiotic not completed by TB patient;
2. Some bacteria survive ; (**R: TB for bacteria**)
3. Spontaneous mutation in bacterial population may produce strains that are resistant to streptomycin ; (**Reject: mutation to give immunity**)
4. Transfer of streptomycin resistance genes from bacterium to bacterium via conjugation/transduction/transformation
5. When streptomycin is applied, it serves as a selection pressure;
6. Bacteria with the resistance gene survive, reproduce and pass the allele to the daughter bacterial cells ;
7. while those that are susceptible/sensitive/ non resistant bacteria die;
8. After a few generations, there is an increase antibiotic resistance allele frequency within the populations of bacteria;

[max 4]

[4]

- (ii) Suggest the mechanism of antibiotic resistance in these resistant strains of *M. tuberculosis* bacteria.

Mutations which led to gene products for e.g.:

- enzymes to breakdown antibiotics
- membrane proteins that inactivate antibiotics
- membrane proteins that pump out antibiotics

[Any 1 for 1 mark]

[1]

- (c) Vaccination has been used successfully to control the spread of many infectious diseases in the population.

Explain how vaccination works in terms of immunological memory and herd immunity.

- Vaccination involves the introduction of a preparation of a dead or weakened form of the disease-causing pathogen (or its antigenic protein) into a person to stimulate the immune system;
- The adaptive immune system is primed to mount a **more rapid and effective response** in a subsequent encounter with the pathogen./ secondary response;
- via the production of a pool **memory B** and **T cells** that carry BCRs that are specific to the antigen in the vaccine and the natural pathogen;
- By vaccinating a critical proportion (**Accept: Idea of, 85% - 95%**) of a community against a contagious disease, susceptible members of a population against an infection/disease can be protected resulting from immunised individuals acting as a barrier against the infection, preventing its spread.

[4]

[Total: 13]